

APLC Approach to Locust Population Management

The APLC's mission is to reduce the credible threat of significant economic damage to agricultural production in a member State (Qld, NSW, Vic and SA) posed by locusts breeding in another member state.

APLC population management programs are aimed at locating and controlling areas of highest density population at the earliest practical opportunity. This strategy is based on intervention during a cycle of population increase, reducing the potential for further increases resulting from migrations and breeding. Control agent use and the total area affected are minimised by only treating areas where locusts are dense enough to be visible from aircraft.

Criteria for APLC intervention

The APLC uses extensive ground surveys, light traps and insect radar to monitor locust activity and development throughout its areas of operation. It also receives and analyses regular reports on locust activity from member states and landholders.

However, for the APLC to be able to initiate timely and efficient intervention to manage significant populations, it must be possible to locate and treat these populations using aircraft.

APLC control programs may be initiated once the total area to be treated is in excess of 10 km² of identified areas, each containing at least 1km² of high density locusts and in close proximity to allow for efficient operations and use of resources.

Medium Density Swarms (>11 adult locusts/m²) and high density bands of nymphs (>1000/m²) that together exceed the 10 km² threshold are considered actionable. At lower densities, the bands of nymphs are usually not visible from the air.

Initiating an APLC control program

All decision to initiate a control programs are consistent with the APLC charter, and are based on known locust biology and population dynamics in relation to climate and habitat conditions.

Numerous factors are considered before initiating a population management intervention, and can include:

- Likelihood of migration – Interstate migration by locusts from certain locations is more prevalent at particular times of year, and may be subject to seasonal and meteorological conditions, such as the development of low pressure trough and frontal weather systems. Locust development and physical condition can also affect the likelihood of a migration event. In situations where locusts are likely to remain at a particular location, they do not present a credible threat to more than one member state and the APLC will not initiate a control program. However, some migrations that

do not present an immediate risk to agriculture could result in a successful breeding cycle that produces a later threat to agricultural areas.

- Developmental stages within the locust life cycle – control programs are designed to reduce locust populations in the areas of highest density. Generally this will occur when the majority of nymphs are at mid-instar stages. Treatment of nymphs is usually more efficient than treatment of adults. Where adult locusts are controlled, the preference is to treat them at the early stages of adult development to reduce the likelihood that breeding or migration has already occurred.
- Location and sensitive areas – where a control program is being conducted by the APLC, significant buffer distances are employed between the areas to be treated and potentially sensitive areas. There are also situations where it is not possible for APLC to apply certain control agents, such as in wetlands, wild life reserves, threatened species habitat, populated areas or in situations where a risk to trade may occur, such as crops approaching harvest or organic farms.
- Human Safety – the safety of all personnel involved in a control program, including landholders, co-operators and contractors are a primary concern for the APLC. No control actions will proceed where a risk to human safety cannot be appropriately managed.
- Logistical considerations – APLC strategic interventions are only undertaken where a significant overall effect can be achieved. Where the situation or the environmental conditions are likely to impact on the level of control which can be achieved, APLC may delay or discontinue the control program.

Prior to implementing control actions on any land, APLC will obtain the informed consent of the landholder or land manager. Without informed consent to apply control agents on the property the APLC will not, under any circumstances, treat the area.

Ceasing APLC control operations

The decision to cease a control program will be made when it is no longer considered efficient or effective to continue, or the risk to people or property has substantially changed.

When the ability to locate treatable areas is reduced relative to the effort to locate those targets, it may no longer be efficient to continue operations at that location. This usually occurs after the highest density part of the locust population has been successfully treated.

In situations where the environmental conditions (such as wind speed or ambient temperature) are likely to impact on the efficacy of the control agent, or where the age structure of the population is likely to reduce efficacy, a control program may be delayed or discontinued. Similarly, factors such as changes in wind direction, or the likely onset of rainfall, which change the potential risk to sensitive areas may result in the program being delayed or discontinued.

At any time where a significant risk to human safety has been identified, control activities will cease immediately until the risk can be managed or the situation substantially changes.

APLC activities should be viewed as a part of a continuum, where action to manage locust outbreaks begins with landholders. Regional authorities and state agencies will also usually be involved in broader population management activities prior to APLC becoming involved in control programs.

Consistent with our mission, APLC population management programs are not intended to provide local crop protection, or district-level control of a locust outbreak, but to prevent losses to agricultural production where a credible inter-state threat exists.

Source: <https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/locusts/landholders/start-stop>